

GRAPH THEORY

Activity 2

Instructions: Answer all problems clearly and completely. Show appropriate diagrams, reasoning, and supporting arguments where necessary. Use a short bondpaper for your solutions. Deadline will be on Thursday, October 2, 2026.

1. River-Crossing Problem.

Illustrate a graph that models a solution to the following problem:

A wolf, a goat, and a cabbage are on one bank of a river. A ferryman wants to take them across, but, since his boat is small, he can take only one of them at a time. For obvious reasons, neither the wolf and the goat nor the goat and the cabbage can be left unguarded. How is the ferryman going to get them across the river?

Your solution should include an appropriate graph representation and a brief explanation of how the graph models the sequence of crossings.

2. Bipartite Graphs and Cycles.

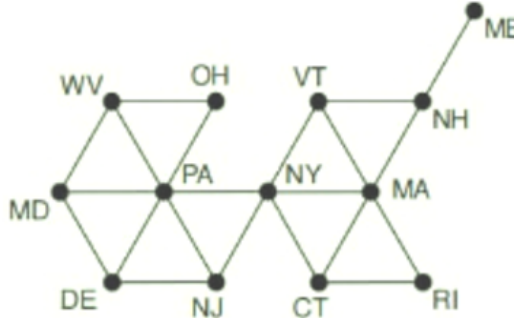
Give examples illustrating the following result:

If a graph is bipartite, then it does not contain an odd cycle. Conversely, if a graph contains an odd cycle, then it is not bipartite.

Provide at least one graph for each case and briefly explain why each graph satisfies the stated property.

3. Walks in a Graph.

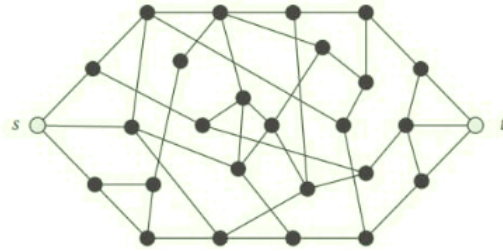
Determine all walks of length 6 that start from Ohio (OH) and end in Pennsylvania (PA) in the graph below.



Clearly list each walk and indicate the sequence of vertices visited. Determine which among these walks is the most efficient. Explain your answer.

4. Shortest Walk.

Consider the graph below. Determine a shortest walk from vertex s to vertex t . Clearly indicate the sequence of vertices in the walk and explain why the walk is a shortest one.

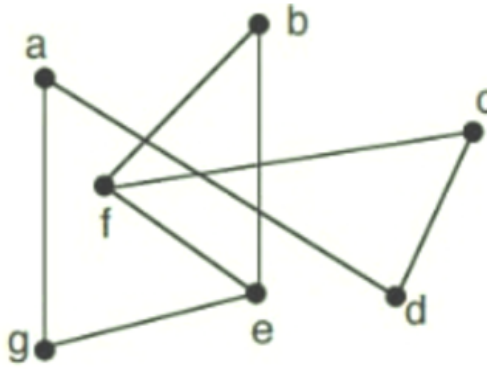


5. Orienting the Roads.

Suppose the graph below represents a network of roads in a certain national park, where the vertices represent intersections and the edges represent roads.

The park officials would like to make every road one-way while maintaining the property that visitors can travel from any intersection to any other intersection.

Is this possible? Justify your answer using an appropriate graph-theoretic argument.



6. Properties of Standard Graphs.

Convention: In this problem set, W_n denotes the wheel graph obtained by joining a new central vertex to every vertex of C_{n-1} . Thus, W_n has n vertices.

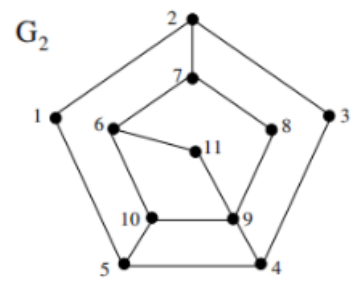
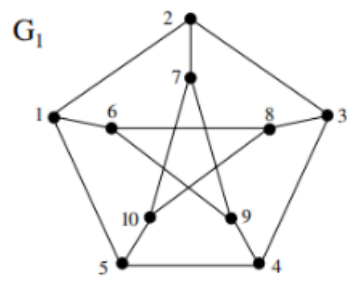
For each of the graphs K_n , P_n , and W_n , complete the following:

- Draw the graph for $n = 4$ and $n = 6$.
- Construct the adjacency matrix for the graph when $n = 5$.
- Determine the following quantities in terms of n :
 - the order;
 - the size;
 - the maximum degree; and
 - the minimum degree.

7. Paths and Cycles of Specified Lengths.

For each of the graphs shown below, determine whether there exists a path of length 9 and a path of length 11. Also determine whether there exists a cycle of length 5, 6, 8, and 9.

For each length, either provide an example by listing the sequence of vertices or explain why such a path or cycle does not exist.



End of Activity 2